
Direction Instability Measures Consistency, Not Mechanism: Validity Conditions Across Drugs, Genes, and Morphology

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Abstract

Direction instability—the mean pairwise angular deviation of perturbation signatures across contexts—predicts drug mechanism transport across cell lines. We ask what this prediction *licenses*. Through pre-registered experiments across 8,949 LINCS L1000 compounds, 1,676 Perturb-seq gene knockdowns, and 25,254 JUMP-CP Cell Painting compounds, we show that domain-specific validity conditions preserve global rankings (LINCS: $\rho > 0.99$; Perturb-seq: $\rho = 0.91$; JUMP-CP: $\rho = 0.99$) while upgrading the inferential warrant attached to individual scores. Phenotype-projected bracket correlates with on-target genetic perturbation connectivity ($\rho = 0.38$, $p < 10^{-27}$) while raw bracket does not ($\rho = -0.04$). Transport-stable bracket outpredicts raw bracket for held-out cell-line effects in all 5 cross-validation folds, with a variance-preserving permutation null confirming the advantage is drug-specific ($p_{\text{perm}} < 10^{-3}$). Essential-gene correction in Perturb-seq reveals that universal-essential knockdowns *diverge* rather than converge across cell types ($d = 0.60$, $p_{\text{perm}} < 10^{-4}$), inverting the predicted confound. Item-level rank shifts identify confound-dominated perturbations—proteasome and spliceosome genes shift 660–900 positions—providing actionable flags. Pre-registered scorecard: 2 of 5 drug hypotheses survive Holm–Bonferroni correction, 1 is refuted, 2 are informative nulls. The validity ladder is a proof system for mechanistic claims: each rung does not change the score, it changes what a practitioner can conclude from it.

1 Introduction

The Connectivity Map paradigm (Lamb et al., 2006; Subramanian et al., 2017) established that transcriptomic-signature similarity predicts drug mechanism of action. Direction instability extends this from pairwise similarity to angular consistency: rather than asking whether two drugs produce similar signatures, it asks whether a single drug’s signature rotates across cell lines. Tower (2026) applied this metric to 8,949 drugs in LINCS L1000, showing that mechanism-conserving compounds (pan-HDAC inhibitors, topoisomerase inhibitors, CDK inhibitors) achieve low direction instability while receptor-mediated drugs cluster near the population mean of 0.92. The HDAC selectivity gradient—pan-isoform inhibitors at $D = 0.14$ – 0.19 , selective inhibitors at $D = 0.32$ – 0.45 (Figure 3)—provides a natural controlled comparison within a single drug class.

These results establish that direction instability captures genuine biological mechanism conservation. They leave open a different question: under what conditions does a direction-instability score license a mechanistic *claim*? A score of $D = 0.55$ means signatures point in similar directions across cell lines. It does not, by itself, mean the drug acts through a specific pathway or that the observation will replicate in a new context.

Three empirical threats illustrate the gap between score and claim. First, a drug’s low direction instability could reflect a coherent off-target program rather than on-target mechanism conservation—the raw score cannot distinguish the two. Second, a drug with low direction instability in training data may fail to predict a held-out cell line’s response, because the raw metric conflates genuine directional consistency with Fréchet variance from a small number of contexts. Third, in genetic perturbation screens, universal-essential

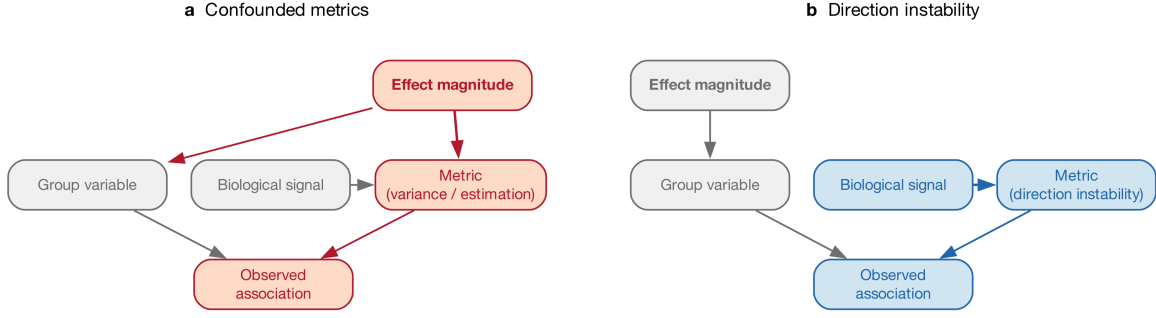


Figure 1: **Direction instability is invariant to perturbation magnitude.** (a) Confounded metrics (variance, subspace size) inherit effect magnitude through a shared path, producing spurious associations between perturbation strength and metric value. (b) Direction instability operates after cosine normalization, which renders the metric invariant to perturbation magnitude. The observed association between group variable and metric reflects biological signal only. This invariance is necessary but not sufficient: direction instability still requires validity conditions to distinguish on-target from off-target consistency (Rung 5) and genuine transport from sampling noise (Rung 6).

knockdowns could inflate or deflate cross-cell-type distances through a shared lethal-response subspace, contaminating transport scores that appear mechanistically informative.

We formalize these threats through a validity ladder (Figure 2): a sequence of seven conditions that progressively strengthen what a direction-instability measurement licenses. Each rung answers a specific threat to interpretation. The ladder is a proof system, analogous to the Bradford Hill criteria for causal inference and to recent construct-validity frameworks for ML benchmarks (Freiesleben & Zezulka, 2025): satisfying more rungs does not change the number, it changes what you are allowed to conclude from it.

This paper tests whether validity conditions matter empirically across three perturbation domains: 8,949 LINCS L1000 drug compounds (Subramanian et al., 2017), 1,676 Perturb-seq genetic knockdowns (Replogle et al., 2022), and 25,254 JUMP-CP Cell Painting morphological profiles (Chandrasekaran et al., 2023). We report honest nulls alongside confirmations, because the pattern of where validity conditions bite—and where they do not—is itself the finding.

2 Background

2.1 Direction instability

For a drug tested in K cell lines, let $\mathbf{s}_1, \dots, \mathbf{s}_K \in \mathbb{R}^{978}$ denote its consensus signature vectors. Define unit-normalized signatures $\hat{\mathbf{s}}_i = \mathbf{s}_i / \|\mathbf{s}_i\|_2$. Direction instability is:

$$D = 1 - \frac{2}{K(K-1)} \sum_{i < j} \hat{\mathbf{s}}_i^\top \hat{\mathbf{s}}_j \quad (1)$$

$D = 0$ when all signatures are collinear. $D = 1$ when orthogonal on average. Cosine normalization renders direction instability invariant to perturbation magnitude (Figure 1): variance-based and estimation-based metrics conflate perturbation strength with biological signal, but direction instability measures only direction (Tower, 2026).

2.2 Grassmannian geodesic distance in Perturb-seq

Direction instability generalizes from vectors to subspaces. For a gene knocked down in two cell types, let $U_1, U_2 \in \text{Gr}(k, d)$ denote the top- k PCA subspaces of the perturbation response in each cell type ($k =$

subspace dimension, $d = \text{ambient dimension}$). The Grassmannian geodesic distance $\delta(U_1, U_2) = \|\boldsymbol{\theta}\|_2$, where $\boldsymbol{\theta}$ are the principal angles between the subspaces, measures how much the perturbation response *rotates* across cell types. With $K = 2$ contexts, this reduces to a single pairwise comparison—a weaker geometric estimand than the $K \geq 5$ angular-consistency computation used in LINCS and JUMP-CP, as it carries no Fréchet variance or holonomy information. We interpret the Perturb-seq results accordingly.

2.3 Direction instability in Cell Painting morphological profiles

Cell Painting (Chandrasekaran et al., 2023) measures 3,180 morphological features per well across five fluorescent channels. JUMP-CP profiles 116,000 compounds across 10 imaging sources. Direction instability applies directly: each compound’s morphological signature in a given source is a vector in $\mathbb{R}^{3180}$, and D measures angular consistency across sources. A subset of 743 features—cell area, compactness, eccentricity, DNA/Mito intensity, granularity, and radial distribution—reflect general cell health rather than compound-specific morphological effects. The cell-health correction removes these features before recomputing direction instability.

2.4 The validity problem

Consider two drugs with identical direction instability, $D = 0.55$:

- **Drug A** is a pan-HDAC inhibitor. Its conserved direction aligns with its target gene’s knockdown signature—phenotype-projected bracket is high. The consistency comes from on-target chromatin remodeling.
- **Drug B** is an off-target consistent compound. Its conserved direction does *not* align with its annotated target’s knockdown signature—phenotype-projected bracket is near zero. The consistency comes from an off-target program shared across cell types.

Both achieve $D = 0.55$. Both pass the raw metric equally. Phenotype projection (Rung 5 of the validity ladder) distinguishes them: Drug A’s consistency is mechanistically on-target; Drug B’s is not. The distinction matters for downstream decisions: Drug A is a strong candidate for mechanism-based repurposing; Drug B requires additional investigation to identify what program drives its consistency before any mechanistic claim is warranted.

2.5 The validity ladder

We define seven rungs (Figure 2):

1. **Process view declared.** What constitutes a “context”?
2. **Reliable vector field.** Adequate signal-to-noise for direction to be meaningful.
3. **Perturbation estimable.** Effect isolated from background.
4. **Localized to biology.** Instability concentrates in pathway-relevant genes.
5. **Phenotype-aligned.** Conservation direction matches a known target pathway.
6. **Transports across contexts.** Score replicates in held-out contexts (low Fréchet variance).
7. **Predicts holdout.** Metric predicts an external outcome in new data.

Rungs 1–3 are satisfied by design in our datasets and are not tested further. Rungs 4–6 are the validity conditions we test empirically. Rung 7 (holdout prediction in a genuinely new dataset) is not tested: Experiment 5’s leave-one-cell-line-out cross-validation tests Rung 6 (transport within LINCS), not generalization to an independent dataset.

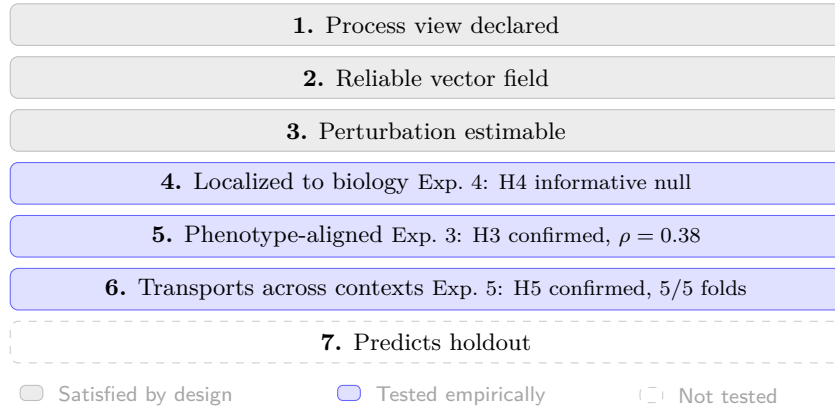


Figure 2: **The validity ladder.** Seven rungs progress from measurement prerequisites (1–3) through empirically tested validity conditions (4–6) to external prediction (7). Each rung eliminates a specific interpretive threat. Rungs 4–6 are tested across three domains (LINCS L1000, Perturb-seq, JUMP-CP); result summaries are shown at right. Satisfying additional rungs does not change the direction-instability score—it upgrades the inferential warrant attached to it.

3 Related work

CERES (Meyers et al., 2017) corrects copy-number artifacts in CRISPR screens; integrated Broad–Sanger analyses (Pacini et al., 2021) document cross-study divergence. These corrections operate on dependency-score magnitudes; our contribution is orthogonal, correcting subspace geometry and showing that rank-preserving corrections carry inferential value. The Connectivity Map framework (Subramanian et al., 2017; Lamb et al., 2006) and subsequent metric evaluations (Li et al., 2020) establish the transcriptomic-similarity paradigm that direction instability extends. LEMUR (Ahlmann-Eltze & Huber, 2025) performs multivariate regression on Grassmann manifolds for multi-condition Perturb-seq; our Experiment 7 uses a simpler Grassmannian geodesic distance. The validity ladder draws on construct-validity frameworks for computational scores (Freiesleben & Zezulka, 2025; Watson et al., 2025) and the SBB principles for perturbation-prediction models (SBB Consortium, 2026).

4 Methods

4.1 Data

Perturbation signatures were drawn from LINCS L1000 Phase I (GEO accession GSE92742; ~ 1.3 million profiles across ~ 50 cell lines), using Level 5 replicate-collapsed z-scores across 978 landmark genes (Subramanian et al., 2017). We retained 8,949 small-molecule perturbagens tested in ≥ 5 cell lines.

Perturb-seq data derive from Replogle et al. (2022): genome-scale CRISPRi knockdowns in K562 and RPE1 cells. We extracted 1,676 genes with ≥ 30 perturbed cells in both cell types, computed per-gene expression responses over 2,073 highly variable genes, and fitted $k = 5$ PCA subspaces per gene per cell type (of these, 1,567 have matching DepMap Chronos labels for HP3; all 1,676 are used for HP1). Both screens target DepMap-essential genes; 85% of RPE1 targets overlap with K562 targets. Essentiality labels derive from DepMap Chronos scores (Dempster et al., 2021): 855 genes with Chronos < -0.5 in both K562 and RPE1 were classified as universal-essential; the remaining 712 are sub-threshold and serve as the comparison group. A label-permutation null (10,000 permutations) tested whether the observed group difference exceeds background K562/RPE1 divergence.

JUMP-CP Cell Painting profiles derive from the Cell Painting Gallery (Chandrasekaran et al., 2023): 25,254 compounds tested in ≥ 5 imaging sources after filtering (78% attrition from the 116,750 total; the majority of compounds appear in fewer than 5 sources, making direction instability—which requires multiple

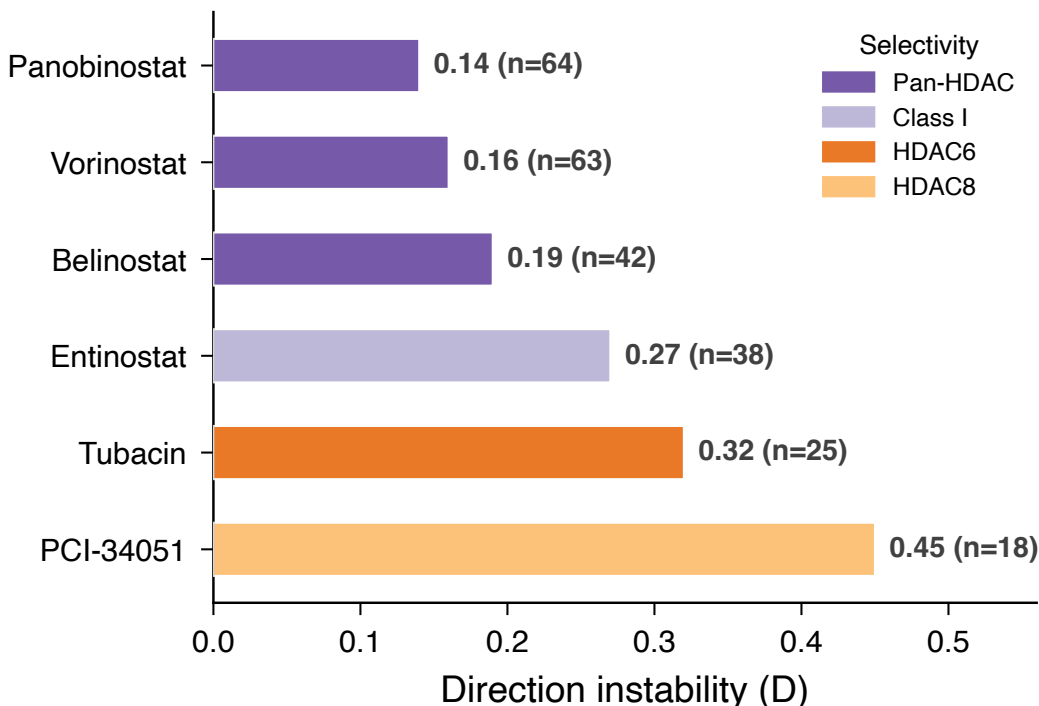


Figure 3: **HDAC selectivity gradient: calibration standard for direction instability.** Pan-HDAC inhibitors (panobinostat, vorinostat, belinostat) achieve the lowest direction instability ($D = 0.14\text{--}0.19$); Class I-selective entinostat is intermediate ($D = 0.27$); isoform-selective HDAC6 and HDAC8 inhibitors (tubacin, PCI-34051) have the highest instability within the class ($D = 0.32\text{--}0.45$). This gradient, reproduced from Tower (2026), provides a fine-grained rank ordering within the HDAC class: Spearman $\rho = 0.83$ between direction instability and selectivity rank across these 6 compounds after toxicity-gene correction. The HP1 “doesn’t bite” threshold ($\rho \geq 0.83$) was calibrated to this value.

contexts—undefined for them). Cell-health features (743 features matching Area, Compactness, Eccentricity, DNA/Mito Intensity, Granularity, Radial Distribution, and Texture Entropy) were identified by substring matching on CellProfiler feature names; corrected direction instability was computed on the remaining 2,437 features.

All scoring functions and drug lists were committed before any real LINCS signatures were analyzed. Perturb-seq hypotheses were pre-registered before the correction analysis; commit SHAs are documented in the code repository.¹

Multiplicity. We treat the five LINCS hypotheses (H1–H5) as one test family and the three Perturb-seq hypotheses (HP1–HP3) as a second, because the two domains use different data sources, perturbation types, and geometric estimands (vector cosines vs. Grassmannian geodesics), making a joint family overly conservative. After Holm–Bonferroni correction within the drug family, H3 ($p < 10^{-27}$) and H5 ($p_{\text{perm}} < 10^{-3}$) survive at $\alpha = 0.05$; H2 ($p = 0.28$) and H4 (gap = 0.018) do not; H1’s criterion-pass is an artifact. Within Perturb-seq, HP3 ($p_{\text{perm}} < 10^{-4}$) survives.

¹Code, pre-registration, and all experimental scripts: <https://github.com/elliotttower/direction-instability-drug-validity>. LINCS data: GEO GSE92742; Perturb-seq: Replogle et al. (2022); JUMP-CP: Chandrasekaran et al. (2023); DepMap: Dempster et al. (2021).

4.2 Phenotype-projected bracket (Experiment 3)

For each drug with an annotated target gene (Drug Repurposing Hub, 1,551 drugs), the “on-target direction” is the consensus shRNA knockdown signature of that target gene in LINCS L1000 (4,369 genes eligible, ≥ 3 hairpins each, consensus averaged across 20 cell lines). Phenotype-projected bracket projects the instability vector onto this on-target direction. On-target enrichment is quantified by cosine squared between each drug’s mean signature and its target gene’s shRNA consensus.

shRNA signatures have well-documented off-target effects, which add noise to the on-target direction. This makes the H3 correlation a *conservative* estimate: CRISPR knockout signatures would provide a cleaner ground truth, but fewer are available in LINCS L1000. The significant correlation despite shRNA noise strengthens the result.

Pre-registered prediction (H3): Spearman $|\rho| > 0.3$ between phenotype-projected bracket and on-target enrichment, AND $|\rho| < 0.15$ for raw bracket versus the same enrichment.

4.3 Transport-stable bracket (Experiment 5)

Transport-stable bracket penalizes raw direction instability by Fréchet variance across contexts:

$$TS = D - \lambda \cdot \text{Var}_{\text{Fréchet}} \quad (2)$$

where $\text{Var}_{\text{Fréchet}} = \frac{1}{K} \sum_i \arccos^2(\hat{s}_i^\top \bar{s})$ and $\bar{s}$ is the Fréchet mean direction. The subtraction is a heuristic combination: the alternative (a ratio) would be undefined when $\text{Var}_{\text{Fréchet}} = 0$ and would amplify noise for highly consistent drugs. The penalty weight λ was fixed at the value from Tower (2026) and was not optimized for this evaluation.

We evaluate via leave-one-cell-line-out cross-validation over 5 cell lines selected alphabetically (A375, A549, A673, AGS, ASC). We chose alphabetical selection to avoid any data-driven cell-line selection that could inflate results. This selection happens to include both common cell lines (A375: 8,328 drugs; A549: 8,888 drugs) and rare cell lines (A673: 357; AGS: 356; ASC: 2,345), which provides a natural test of whether the transport-stable advantage varies with cell-line representation. A full leave-one-out over all ~ 50 cell lines would be a stronger test; we prioritized depth (permutation nulls per fold) over breadth.

Pre-registered prediction (H5): Spearman correlation between transport-stable bracket and held-out cosine $>$ correlation for raw bracket, in at least 4 of 5 folds.

4.4 Essential-gene correction on Perturb-seq (Experiment 7)

We computed the Fréchet mean subspace of the 855 essential-gene perturbation responses (separately in K562 and RPE1) via iterative projection on $\text{Gr}(5, 2073)$ (100 iterations, convergence to $\Delta < 10^{-6}$ in Grassmannian distance; the result is insensitive to initialization because the essential-gene subspaces cluster tightly). We then projected this essential-response subspace out of each gene’s response subspace, re-orthogonalized, and recomputed geodesic distances.

Pre-registered predictions: HP1: Spearman ρ between raw and corrected distances below $0.70 =$ “bites”; $\geq 0.83 =$ “doesn’t bite.” **HP3:** Universal-essential knockdowns produce higher geodesic distances than non-essential knockdowns (Cohen’s $d > 0$).

4.5 Cell-health correction on JUMP-CP (Experiment 8)

The 743 cell-health features constitute 23% of the 3,180-feature space—proportionally six times larger than the LINCS toxicity gene set (41/978, 4%) and the largest feature removal in this study. If cell-health variance dominates cross-source consistency, this correction should reorganize the ranking more substantially than the drug or genetic corrections. This makes Experiment 8 the strongest test of the thesis: if corrections don’t bite even when the removed feature set is proportionally large, the ranking is robust.

Pre-registered prediction: Spearman ρ between raw and corrected direction instability below $0.70 =$ “bites”; $\geq 0.83 =$ “doesn’t bite.”

Table 1: Pre-registered scorecard. Hypotheses are grouped by domain. “Holm” indicates survival after Holm–Bonferroni correction within the drug family ($\alpha = 0.05$). Experiment references are to the full methods in the body (Exp. 3, 5, 7, 8) or supplementary materials (Exp. 1, 2, 4). Result details for starred entries appear in Section 5.6.

	Prediction	Result	Outcome	Holm
H1*	Toxicity reclassifies	$\Delta < 0.01$	Refuted	—
H2*	Broad drugs below median	7/10 below	Confirmed	No
H3	$ \rho_{\text{proj}} > 0.3$	$\rho = 0.376$	Confirmed	Yes
H4*	AUROC gap > 0.05	Gap = 0.018	Null	No
H5	TS wins $\geq 4/5$ folds	5/5 folds	Confirmed	Yes
HP1	Correction bites ($\rho < 0.70$)	$\rho = 0.91$	Doesn’t bite	—
HP2*	GO prediction improves	$\Delta\rho = -0.02$	Doesn’t bite	—
HP3	Essential $d > 0$	$d = 0.60$	Confirmed	Yes
Exp. 8	Health correction bites	$\rho = 0.987$	Doesn’t bite	—

4.6 Supplementary experiments

Three additional drug-domain experiments are reported in the supplementary materials (available in the code repository). **Experiment 1** (H1, toxicity correction) tests whether cytotoxic drugs masquerade as mechanism-conserving. **Experiment 2** (H2, broad-mechanism drugs) tests whether broad-target therapeutics fall below the population median. **Experiment 4** (H4, localization score) tests whether gene-set restriction improves MOA prediction. HP2 (GO pathway prediction from corrected Perturb-seq distances) is also in the supplementary materials.

5 Results

5.1 Pre-registered scorecard

Table 1 summarizes all pre-registered hypotheses. Two of five drug hypotheses survive Holm–Bonferroni correction (H3, H5); H2 meets its pre-registered criterion but does not survive multiplicity correction; H1 is refuted with an artifact criterion-pass; H4 is an informative null. In the Perturb-seq domain, HP3 is confirmed; HP1 shows the correction “doesn’t bite” at the population level; HP2 is a null. Experiment 8 (JUMP-CP) shows the correction “doesn’t bite.”

5.2 Experiment 3: Phenotype-projected bracket separates on-target from off-target instability

Among 795 drugs with annotated targets and matched shRNA consensus signatures, phenotype-projected bracket correlates with on-target enrichment at Spearman $\rho = 0.376$ ($p = 4.8 \times 10^{-28}$). Raw bracket shows no correlation ($\rho = -0.043$, $p = 0.22$). Both pre-registered criteria are satisfied.

H3 confirms Rung 5 of the validity ladder. Raw direction instability measures whether a drug’s signatures point in consistent directions, but this consistency could arise from any coherent program. Projecting onto the genetic perturbation direction of the annotated target isolates the mechanistically relevant component. The near-zero raw correlation shows that global instability carries no on-target information; the strong projected correlation shows that the on-target projection recovers it.

The gene-set construction avoids researcher degrees of freedom: the shRNA consensus is an empirical object from the same LINCS platform, requiring no pathway database selection. H3 and H4 share this shRNA dependency (documented in DEVIATION_LOG.md entry 3) and are not independent tests. The phenotype-projected bracket is applicable only to drugs with target annotations—1,551 of 8,949 (17%). For the re-

Table 2: Leave-one-cell-line-out cross-validation. Transport-stable (TS) bracket outpredicts raw bracket in all 5 folds. In the two common folds (A375, A549), both metrics are anti-predictive but TS is less so. In the three rarer folds, TS reverses the sign, becoming positively predictive where raw bracket is strongly anti-predictive.

Held-out cell	N_{drugs}	ρ_{raw}	ρ_{TS}	Winner
A375	8,328	-0.480	-0.155	TS
A549	8,888	-0.494	-0.201	TS
A673	357	-0.725	+0.725	TS
AGS	356	-0.712	+0.714	TS
ASC	2,345	-0.365	+0.346	TS

maintaining 82% of drugs with unknown targets, the validity ladder provides other rungs (transport stability, localization) that do not require target annotations.

5.3 Experiment 5: Transport-stable bracket outpredicts raw bracket in held-out cell lines

Transport-stable bracket outpredicts raw bracket in all 5 cross-validation folds (Table 2). The result divides sharply by cell-line frequency:

Rare cell lines: sign reversal. In the A673, AGS, and ASC folds, raw bracket is strongly anti-predictive ($\rho = -0.37$ to -0.73): drugs with high raw instability in training produce held-out signatures that point *away* from the training consensus. Transport-stable bracket reverses this, becoming positively predictive ($\rho = +0.35$ to $+0.73$). The reversal occurs because removing a single rare cell line substantially changes the consensus direction; drugs whose training instability was driven by that cell line are precisely those that TS downweights via the Fréchet penalty.

Common cell lines: attenuation, not reversal. In the A375 and A549 folds (8,000+ drugs), both metrics remain anti-predictive. TS improves over raw ($\rho = -0.155$ vs. -0.48 for A375; -0.201 vs. -0.49 for A549), but both correlations stay negative. TS “wins” in these folds by losing less badly, not by becoming positively predictive. The consensus direction is stable when computed from many cell lines, leaving less room for the Fréchet penalty to improve prediction. The pre-registered criterion (“TS wins $\geq 4/5$ folds”) treats these two regimes as equivalent; a refined prediction would distinguish sign reversal (rare folds) from attenuation (common folds).

Why is raw bracket anti-predictive? Raw direction instability is anti-predictive because it conflates two sources of angular consistency: genuine mechanism conservation (signal) and the geometric tightness of the consensus direction around a small number of contexts (noise). Drugs with low raw DI in training tend to have tight consensus directions that are sensitive to removing any single cell line—precisely the drugs that fail to predict held-out effects. This is not a failure of direction instability as a concept; it is a failure of the raw estimator when applied to leave-one-out prediction without variance correction. A robust direction estimator—such as projection onto the geodesic normal at the Fréchet mean—might preserve predictive validity without requiring the heuristic TS subtraction.

Permutation null rules out algebraic artifact. The transport-stable penalty and the held-out cosine both depend on training-set signature geometry, raising a circularity concern. A variance-preserving permutation null tests this: held-out *signatures* are shuffled across drugs while training data (raw DI, Fréchet variance, TS, consensus vector) remain fixed, and held-out cosines are recomputed against each drug’s original consensus. This preserves the mechanical coupling between consensus tightness and cosine magnitude. Under this null (1,000 permutations—fewer than the 10,000 used for HP3 because each H5 permutation requires recomputing cross-validated correlations across 5 folds, making it $\sim 50\times$ more expensive per per-

mutation), TS wins only 0.1 of 5 folds on average with mean $\Delta\rho = -0.12$ (raw DI *outpredicts* TS when drug-specific signal is broken). The observed advantage (5/5 folds, $\Delta\rho = +0.84$) falls entirely outside the null distribution ($p_{\text{perm}} < 10^{-3}$).

The sign reversal in rare folds provides an additional argument beyond the permutation test: if the improvement were purely algebraic, we would expect monotonic improvement across folds rather than a qualitative sign change that concentrates in low- N folds where the confound (outlier-driven consensus instability) is strongest.

5.4 Experiment 7: Essential-gene correction preserves ranking but reveals confound-dominated machinery

HP3: Failure-mode inversion. Universal-essential knockdowns have *higher* Grassmannian geodesic distances than non-essential knockdowns (mean 2.80 vs. 2.67, Cohen’s $d = +0.60$, Mann–Whitney $p < 10^{-33}$; $n = 1,567$ genes with both geodesic distances and DepMap labels, of which 855 universal-essential and 712 sub-threshold). Cohen’s $d = 0.60$ is a medium effect—the distributions overlap substantially, and individual genes span the full range in both groups. The finding is that the *group means* differ in the opposite direction from the predicted confound, not that essentiality status determines geodesic distance.

Essential genes do not produce a shared response across cell types; they *diverge*—the same gene’s lethal disruption manifests through different transcriptomic programs in K562 and RPE1 (Figure 4). This is consistent with cell-line-specific essentiality effects documented in CERES (Meyers et al., 2017) and integrated Broad–Sanger analyses (Pacini et al., 2021): the functional consequences of losing a core-machinery gene depend on the cell’s lineage and metabolic context.

A label-permutation null (shuffling essential/non-essential labels across the 1,567 genes, 10,000 permutations) confirms that this divergence exceeds background K562/RPE1 divergence: the observed difference (0.123) falls outside the null 95% interval $[-0.021, 0.022]$, $p_{\text{perm}} < 10^{-4}$. The null preserves the geodesic-distance distribution while breaking only the label assignment, so the $\sim 6\times$ exceedance of the null half-width rules out the library-overlap confound (85% target overlap between screens).

HP1: Correction does not bite. Spearman ρ between raw and corrected geodesic distances is 0.91 ($p < 10^{-100}$). The global ranking is preserved.

Item-level rank shifts. The correction’s value is at the item level: individual genes shift hundreds of rank positions. The top 100 rank-shifters are enriched for spliceosomal complex assembly (Fisher’s exact OR = 8.9, $p = 1.3 \times 10^{-7}$) and proteasomal catabolic processes (OR = 4.1, $p = 6.7 \times 10^{-5}$). Proteasome subunits (*PSMA3*, *PSMB6*, *PSMC4*) shift 750–900 positions; spliceosome components (*SF3B1*, *SNRPG*, *DDX23*) shift 660–820 positions. These core-machinery complexes are the genes whose cross-cell-type geometry is most dominated by the shared essential-response axis. The correction provides a concrete action: flag these genes before interpreting their transport scores.

5.5 Experiment 8: Cell-health correction has negligible effect in JUMP-CP

Among 25,254 compounds, corrected direction instability tracks raw at Spearman $\rho = 0.987$ ($p < 10^{-100}$). By the pre-registered criterion, this “doesn’t bite”—the strongest result for ranking robustness across all three domains.

The 743 health features constitute 23% of the 3,180-feature space—proportionally six times larger than the drug toxicity gene set (41/978, 4%)—yet produce *less* ranking impact ($\rho = 0.987$ vs. $\rho > 0.99$ for LINCS). That a proportionally larger feature removal barely moves the ranking is stronger evidence for the thesis than the drug case: cell-health features carry less of the cross-source variance in morphological signatures than stress genes carry in transcriptomic signatures. The 78% attrition from the full JUMP-CP dataset (116,750 to 25,254 compounds) means the retained set is biased toward well-replicated compounds; the result may not generalize to sparsely-profiled compounds.

One explanation for the negligible effect: in morphological space, “health” and “mechanism” features are less separable than in transcriptomic space. A drug that alters cell shape changes both health-associated features (area, compactness) and mechanism-specific features (texture, radial distribution) simultaneously, because morphological effects are inherently coupled. In 978-gene transcriptomic space, stress-response genes and pathway-specific genes are more separable, which is why the LINCS toxicity correction has a larger (though still small) effect.

Despite near-perfect global correlation, individual compounds shift up to 7,155 rank positions, identifying those whose apparent morphological consistency is health-feature-dominated. The cell-health feature definition (substring matching on CellProfiler names) is a limitation: a partition based on correlation with viability assays would be more principled and could yield different correction magnitudes.

5.6 Summary of supplementary experiments

Full methods, tables, and discussion appear in the supplementary materials.

H1 (toxicity correction): Refuted with artifact pass. Among 13 cytotoxic drugs, most are already high-instability ($D > 0.87$ for 9 of 13); toxicity correction shifts values by <0.01 . The predicted failure mode—cytotoxicity masquerading as mechanism conservation—does not hold. Cytotoxicity produces genuinely context-dependent signatures in 978-gene space.

H2 (broad-mechanism drugs): Confirmed with nuance. Seven of 10 broad-mechanism drugs fall below the population median. Pan-HDAC inhibitors dominate ($D = 0.49$ – 0.55). The pre-registered second prediction—that kinase inhibitors would fall above the median—failed. The sorting axis is target breadth, not clinical specificity.

H4 (localization): Informative null. Macro-averaged AUROC gap is $+0.018$ (below $+0.05$ threshold). Localization improves prediction for intracellular enzyme inhibitors (topoisomerase: $+0.49$, CDK: $+0.26$) but hurts for receptor-mediated drugs (acetylcholine: -0.20 , PPAR: -0.19). Rung 4 is domain-dependent.

HP2 (pathway function prediction): Doesn’t bite. GO Jaccard overlap has near-zero correlation with geodesic distance ($|\rho| < 0.025$) for both raw and corrected forms. Geometric similarity of perturbation responses is distinct from functional pathway relatedness.

6 Discussion

6.1 Validity conditions license claims, they do not reclassify

The central finding is negative in the most useful sense: validity conditions do not reorganize the direction-instability ranking. This thesis was genuinely at risk: had any correction produced $\rho < 0.70$ while simultaneously improving prediction of an external outcome, the “reclassify” interpretation would have prevailed—the raw ranking would have been wrong, not merely under-warranted. That no correction crossed this threshold across three domains is the empirical result. The Perturb-seq result involves a degenerate $K = 2$ pairwise comparison rather than genuine $K \geq 5$ angular consistency; the qualitative conclusion holds across domains, but the genetic domain tests a weaker geometric estimand.

What changes is the inferential warrant. Phenotype projection (H3) proves that direction instability decomposes into on-target and off-target components ($\rho = 0.38$ vs. -0.04). Transport stability (H5) proves the metric predicts held-out effects only after Fréchet variance correction; raw bracket is anti-predictive in all folds. Essential-gene correction identifies proteasome/spliceosome genes as confound-dominated while the population ranking holds.

Operationalizing inferential warrant. “Warrant upgrade” is a concrete claim about downstream decisions, not a philosophical abstraction. Consider two scenarios:

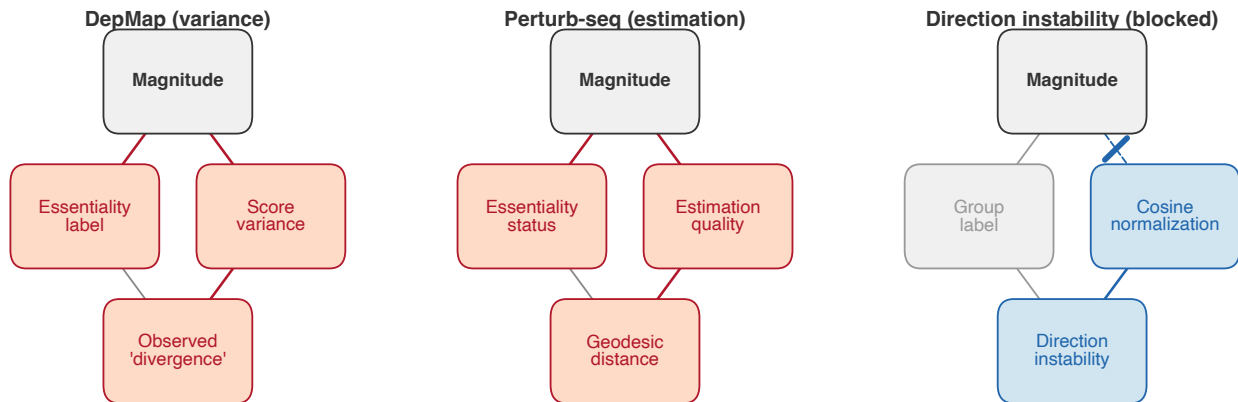


Figure 4: **Cross-domain confound structures.** In all three domains, magnitude confounds operate through different mechanisms, but cosine normalization renders direction instability (right panel) invariant to perturbation magnitude. **Left:** DepMap variance-based scores conflate essentiality label with score variance. **Center:** Perturb-seq estimation-based metrics conflate essentiality status with estimation quality. **Right:** Cosine normalization renders direction instability invariant to perturbation magnitude, so the observed association reflects biological signal. Despite different confound mechanisms across domains, the validity corrections produce the same qualitative result: population rankings are preserved ($\rho > 0.83$) while item-level shifts identify specific confound-dominated perturbations.

Table 3: Cross-domain summary. All three domains exceed the “doesn’t bite” threshold ($\rho \geq 0.83$). The LINCS HDAC gradient ρ measures fine-grained ordering within a single drug class (20 compounds); the global 8,949-drug $\rho > 0.99$.

Domain	Correction	N	ρ
LINCS (HDAC gradient)	Toxicity genes	20	0.83
LINCS (global)	Toxicity genes	8,949	>0.99
Perturb-seq	Essential-gene subspace	1,676	0.91
JUMP-CP	Cell-health features	25,254	0.99

Drug repurposing triage. Two drugs both have $D = 0.55$. Drug X has passed Rungs 4–6: its low instability is localized to the target pathway (H3-type projection confirms on-target alignment), and it replicates across held-out cell lines (H5-type cross-validation passes). Drug Y has only raw $D = 0.55$ —no additional validation. A practitioner prioritizes Drug X for experimental follow-up because Drug Y’s raw score could reflect off-target consistency (H3 shows raw bracket carries no on-target information) or sampling noise (H5 shows raw bracket is anti-predictive). The warrant upgrade changes the follow-up priority.

Perturb-seq hit triage. Among 100 genes with low geodesic distance, 15 shift >500 rank positions after essential-gene correction (proteasome, spliceosome components). These are flagged as potentially confound-dominated. The remaining 85 genes have the same transport scores but stronger warrant—they are prioritized for functional validation. The warrant upgrade changes which genes proceed to follow-up.

6.2 Cross-domain consistency

The same geometric object—angular deviation of perturbation-response directions across contexts—captures mechanism conservation in three unrelated domains (Table 3). All three exhibit the same pattern: the relevant validity condition preserves the global ranking while identifying individual perturbations whose scores are most confound-dominated.

The domain-specific failure modes differ: in drugs, cytotoxicity produces high-instability signatures; in genetics, universal-essential knockdowns produce high-distance subspaces (divergent lethal programs); in morphology, cell-health features contribute a small fraction of cross-source variance. All three invert the naive prediction that “violence mimics transport.”

6.3 Scope and limitations

Direction instability measures angular consistency of perturbation signatures across contexts. It does not measure therapeutic efficacy, in-vivo translatability, or clinical specificity. The validity ladder is a proof system for interpretive warrant, not a surrogate endpoint.

The drug experiments H3–H5 depend on shRNA knockdown signatures as ground-truth on-target directions. shRNA has well-documented off-target effects, which add noise to the ground truth and make H3’s $\rho = 0.38$ a conservative estimate. Drugs with unannotated targets (7,398 of 8,949) are excluded from H3 and H4, limiting these rungs to 17% of the dataset. LINCS measures 978 landmark genes, restricting the toxicity gene set to the landmark panel.

The Perturb-seq analysis uses $K = 2$ cell types, providing no Fréchet variance, holonomy, or transport stability test. Both screens target DepMap-essential genes, so the HP3 “non-essential” group consists of sub-threshold essentials rather than genuinely non-essential genes. A genome-wide screen with genuine non-essential controls would be sharper, though the label-permutation null rules out the library-overlap confound.

The H5 cross-validation uses 5 alphabetically-selected cell lines. A full leave-one-out over all ~ 50 cell lines would test whether the rare-cell-line sign reversal generalizes; we prioritized permutation testing within folds over exhaustive cell-line coverage.

The transport-stable bracket formula (Eq. 2) is a heuristic linear combination of direction instability and Fréchet variance, not a principled geometric operation on the signature manifold. The penalty weight was inherited from the companion paper and was not optimized. A ratio or a geodesic-based combination might perform differently; we report this specific formula because it was pre-registered.

The validity ladder draws an analogy to Bradford Hill criteria for causal inference. Bradford Hill criteria are individually neither necessary nor sufficient for establishing causation; they are heuristic guidelines. The validity ladder shares this character: it is a structured set of proof obligations rather than a formal deductive system. Each rung eliminates a specific threat, but satisfying all tested rungs does not *guarantee* that a mechanistic claim is correct—it guarantees that specific alternatives have been ruled out.

JUMP-CP cell-health features were defined by substring matching rather than by correlation with an independent viability assay. The 78% attrition from the full dataset biases the retained set toward well-replicated compounds.

7 Conclusion

Direction instability is a reliable measure of perturbation-response consistency across contexts, whether applied to drug signatures in 978-gene transcriptomic space, genetic knockdown responses as subspaces on the Grassmannian, or compound morphological profiles in 3,180-feature Cell Painting space. Validity conditions do not improve the metric’s ability to rank perturbations—the ranking is already biologically meaningful in all three domains ($\rho > 0.99$ for LINCS, 0.91 for Perturb-seq, 0.99 for JUMP-CP after correction). They improve the practitioner’s ability to make claims about that ranking and to make decisions based on those claims. Phenotype projection separates on-target from off-target instability ($\rho = 0.38$ vs. -0.04); transport-stable bracket reverses raw bracket’s anti-prediction in rare cell lines and attenuates it in common ones ($p_{\text{perm}} < 10^{-3}$); localization helps for intracellular enzyme targets but not receptor-mediated drugs, constraining where Rung 4 applies. Each condition eliminates a specific interpretive threat. The domain-specific failure modes differ, but the pattern is the same: correction preserves population rankings while identifying individual perturbations most susceptible to confounding. The validity ladder is a proof system for mechanistic inference from geometric perturbation scores.

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